

COURSE CODE

SBT-DF203

COURSE TITLE

Basic Networking Skills for Digital Forensics

LAB TITLE

SMTP Email Traffic Forensics

NAME

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STUDENT ID

2025/FWSD/11325

DATE

15/09/2026

PLATFORM USED

Kali Linux

INSTRUCTOR

Mr. Aminu Idris, AMCPN
Commandant

Executive Summary

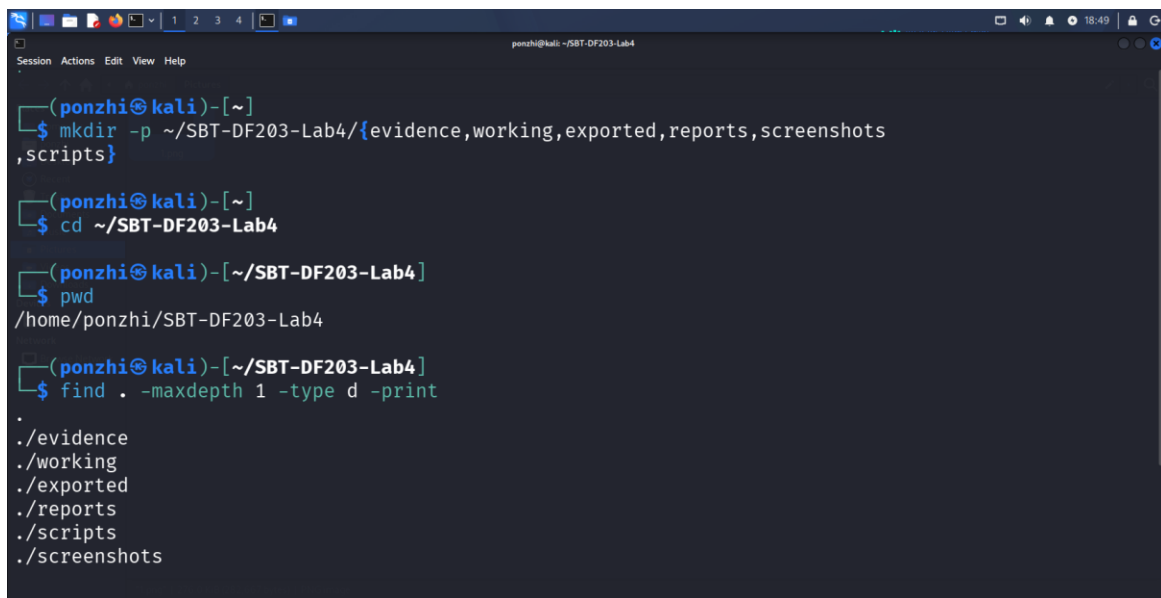
This report details the forensic analysis of a historical email packet capture (smtp.pcap), which is a publicly available sample from Wireshark used for training. The capture shows a plaintext SMTP session between a Microsoft Office Outlook 12.0 client (10.10.1.4) and a mail server at xc90.websitewelcome.com (74.53.140.153), recorded on October 5, 2009.

The analysis recreated the SMTP command and response exchange. It also decoded the Base64-encoded AUTH LOGIN credentials, recovered the email message headers and body, and identified metadata such as client and server information, plus network details (IP addresses, ports, and MAC addresses). Even though the server supported STARTTLS, the client did not use it, resulting in the entire session—including the authentication exchange and message content—being sent in plaintext.

All original evidence was preserved and verified with hashes before analysis. The analysis itself took place on a verified working copy in an isolated lab environment, meeting the evidence-handling requirements for this task.

Lab Folder Structure and Evidence Preparation

The lab folder structure was set up before handling any evidence. The SMTP sample capture was copied into a working directory, while the original file was kept unchanged in the evidence folder.



```
ponzhi@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help

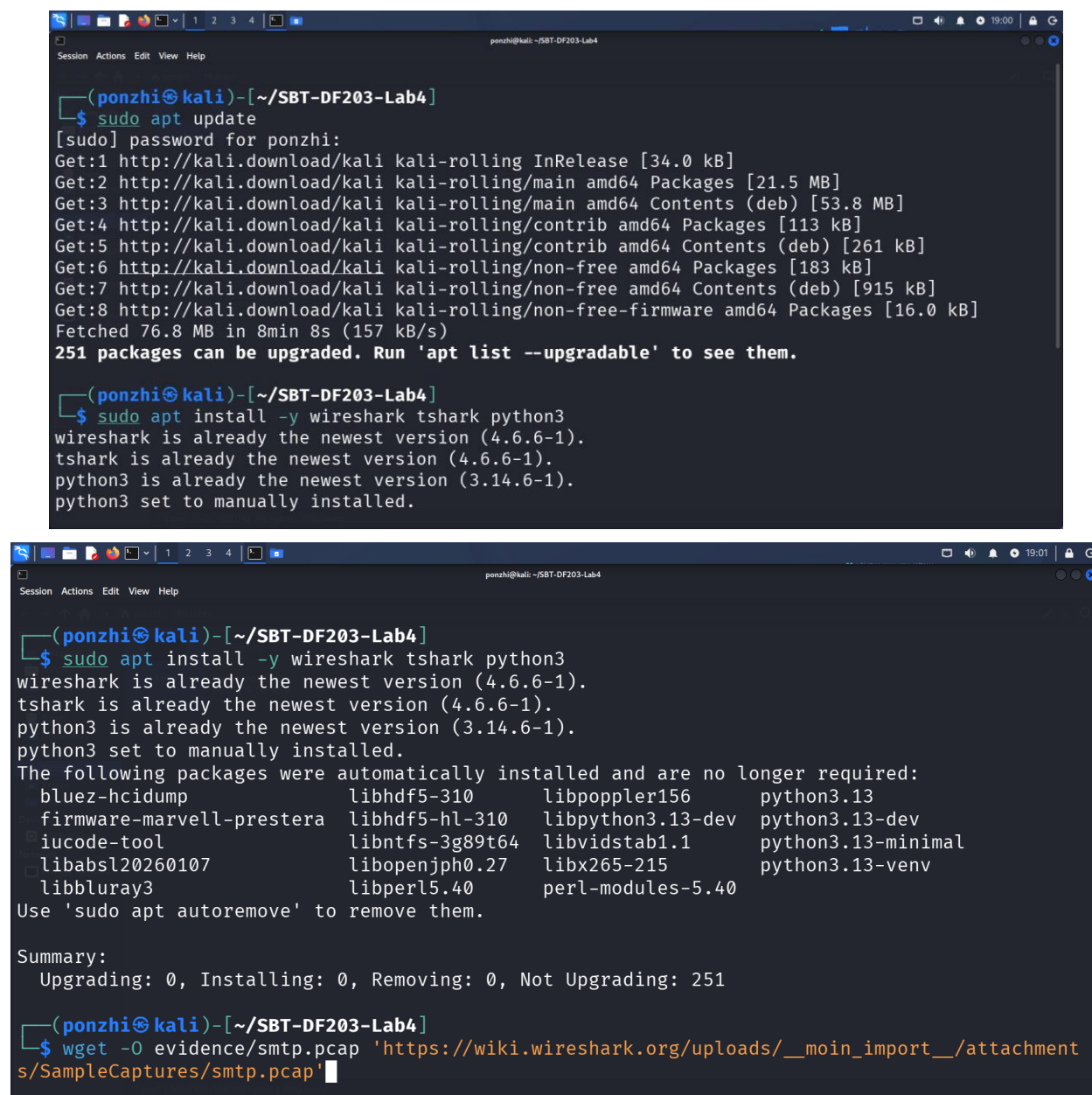
(ponzhi@kali)-[~]
$ mkdir -p ~/SBT-DF203-Lab4/{evidence,working,exported,reports,screenshots,scripts}

(ponzhi@kali)-[~]
$ cd ~/SBT-DF203-Lab4

(ponzhi@kali)-[~/SBT-DF203-Lab4]
$ pwd
/home/ponzhi/SBT-DF203-Lab4

(ponzhi@kali)-[~/SBT-DF203-Lab4]
$ find . -maxdepth 1 -type d -print
.
./evidence
./working
./exported
./reports
./scripts
./screenshots
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working



```
(ponzhi@kali)-[~/SBT-DF203-Lab4]
$ sudo apt update
[sudo] password for ponzhi:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.5 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.8 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [113 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [261 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [16.0 kB]
Fetched 76.8 MB in 8min 8s (157 kB/s)
251 packages can be upgraded. Run 'apt list --upgradable' to see them.

(ponzhi@kali)-[~/SBT-DF203-Lab4]
$ sudo apt install -y wireshark tshark python3
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
python3 is already the newest version (3.14.6-1).
python3 set to manually installed.

(ponzhi@kali)-[~/SBT-DF203-Lab4]
$ sudo apt install -y wireshark tshark python3
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
python3 is already the newest version (3.14.6-1).
python3 set to manually installed.
The following packages were automatically installed and are no longer required:
  bluez-hcidump      libhdf5-310      libpoppler156     python3.13
  firmware-marvell-presteria libhdf5-hl-310   libpython3.13-dev  python3.13-dev
  iucode-tool        libntfs-3g89t64  libvidstab1.1     python3.13-minimal
  libabsl20260107    libopenjph0.27   libx265-215       python3.13-venv
  libbluray3         libperl5.40      perl-modules-5.40
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 251

(ponzhi@kali)-[~/SBT-DF203-Lab4]
$ wget -O evidence/sntp.pcap 'https://wiki.wireshark.org/uploads/__moin_import__/attachment
s/SampleCaptures/sntp.pcap'
```

```
ponzhi@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help
Resolving wiki.wireshark.org (wiki.wireshark.org) ... 104.26.10.240, 172.67.75.39, 104.26.11.240, ...
Connecting to wiki.wireshark.org (wiki.wireshark.org)|104.26.10.240|:443 ... connected.
HTTP request sent, awaiting response ... 302 Found
Location: https://gitlab.com/wireshark/wireshark/-/wikis/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap [following]
--2026-09-15 19:01:20-- https://gitlab.com/wireshark/wireshark/-/wikis/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap
Resolving gitlab.com (gitlab.com) ... 172.65.251.78, 2606:4700:90:0:f22e:fbec:5bed:a9b9
Connecting to gitlab.com (gitlab.com)|172.65.251.78|:443 ... connected.
HTTP request sent, awaiting response ... 200 OK
Length: 27850 (27K) [application/octet-stream]
Saving to: 'evidence/smtp.pcap'

evidence/smtp.pcap      100%[=====>] 27.20K  --.-KB/s    in 0.1s

2026-09-15 19:01:27 (214 KB/s) - 'evidence/smtp.pcap' saved [27850/27850]

(ponzhi@kali)-[~/SBT-DF203-Lab4]
$ clear
```

We hashed the original evidence file and its working copy using SHA-256 right after we collected them. We also used capinfos to record capture details for the chain of custody.

```
(ponzhi@kali)-[~/SBT-DF203-Lab4]
$ cp --preserve=timestamps evidence/smtp.pcap working/smtp_working.pcap
sha256sum evidence/smtp.pcap working/smtp_working.pcap | tee reports/smtp_capture_hashes.txt
capinfos evidence/smtp.pcap | tee reports/smtp_capinfos.txt
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab  evidence/smtp.pcap
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab  working/smtp_working.pcap
File name:          evidence/smtp.pcap
File type:          Wireshark/tcpdump/ ... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit:  file hdr: 65535 bytes
Number of packets:  60
File size:          27 kB
Data size:          26 kB
Capture duration:   9.198384 seconds
Earliest packet time: 2009-10-05 07:06:07.492060
Latest packet time:  2009-10-05 07:06:16.690444
Data byte rate:     2,920 bytes/s
Data bit rate:      23 kbps
Average packet size: 447.77 bytes
```

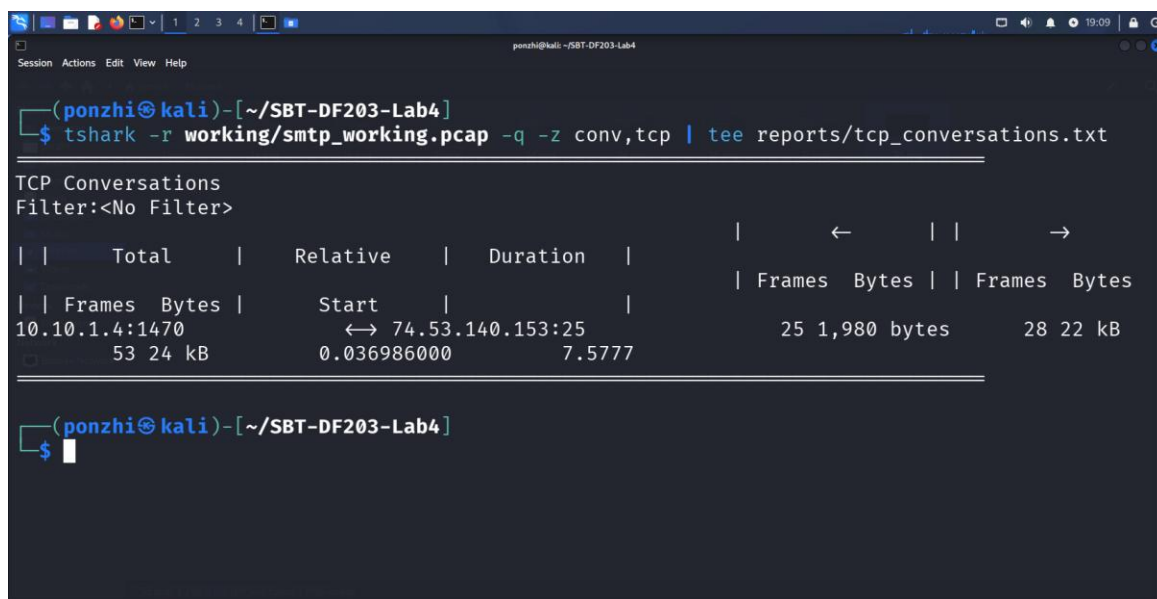
Both the original and working copy produced an identical SHA-256 hash (17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab), confirming the working copy is the same, unaltered.

Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab4-Kumpon_Emmanuel_Nandom
Trainee name	Kumpon Emmanuel Nandom
Date and time started	15 September 2026, at about 6:49 pm
Evidence file name(s)	evidence/smtp.pcap (original), working/smtp_working.pcap (working copy)
Source or generation method	Downloaded via wget from the Wireshark wiki public sample-captures repository, as instructed
Original SHA-256	17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
Working-copy SHA-256	17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab (matches original)
Analysis workstation/VM	Kali Linux virtual machine, user ponzhi@kali, isolated lab network
Notes on any changes	None. Working copy hash matches the original; all analysis performed on the working copy only, original left untouched in evidence/

Part A - Inventory the Capture and Locate SMTP Streams

The capture was inventoried to confirm the TCP conversation and locate SMTP traffic before deeper analysis.



```
ponzhi@kali: ~/SBT-DF203-Lab4
$ tshark -r working/smtp_working.pcap -q -z conv,tcp | tee reports/tcp_conversations.txt

TCP Conversations
Filter:<No Filter>

| | Total | Relative | Duration | | | | |
| | Frames Bytes | Start | | Frames Bytes | | Frames Bytes |
10.10.1.4:1470 <-> 74.53.140.153:25 25 1,980 bytes 28 22 kB
53 24 kB 0.036986000 7.5777

(ponzhi@kali) - [~/SBT-DF203-Lab4]
$
```

The client connected using temporary port 1470, and the server replied on the standard SMTP port 25. The entire SMTP session lasted from October 5, 2009, at 07:06:07.492 to 07:06:16.690, which is about 9.2 seconds. During this time, 25 important SMTP frames were identified using the SMTP display filter.

Command/Response Analysis

I used tshark to capture the full SMTP request and response exchange. This included the smtp.req.command, smtp.req.parameter, smtp.response.code, and smtp.rsp.parameter fields.

```
ponzhi@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help

(ponzhi@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'smtp.req || smtp.rsp' -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e smtp.req.command -e smtp.req.parameter \
-e smtp.response.code -e smtp.rsp.parameter \
| tee reports/smtp_commands_responses.tsv
6      2009-10-05T07:06:08.219663000+0100      74.53.140.153      10.10.1.4      2
20      xc90.websiteswelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 ,We do not
      authorize the use of this system to transport unsolicited, ,and/or bulk e-mail.
7      2009-10-05T07:06:08.224809000+0100      10.10.1.4      74.53.140.153      EHLO      GP
9      2009-10-05T07:06:08.566183000+0100      74.53.140.153      10.10.1.4      2
50      xc90.websiteswelcome.com Hello GP [122.162.143.157],SIZE 52428800,PIPELINING,AUTH PLAI
N LOGIN,STARTTLS,HELP
10      2009-10-05T07:06:08.568729000+0100      10.10.1.4      74.53.140.153      AUTH      LOGIN
11      2009-10-05T07:06:08.911081000+0100      74.53.140.153      10.10.1.4      3
34      VXNlcm5hbWU6
12      2009-10-05T07:06:08.911655000+0100      10.10.1.4      74.53.140.153
13      2009-10-05T07:06:09.253544000+0100      74.53.140.153      10.10.1.4      3
34      UGFzc3dvcmQ6
14      2009-10-05T07:06:09.254118000+0100      10.10.1.4      74.53.140.153
```

```
ponzhi@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help

: 334 UGFzc3dvcmQ6
14      2009-10-05T07:06:09.254118000+0100      10.10.1.4      1470      74.53.140.153      25      C
: Pass: cHVuamFiQDEyMw==
15      2009-10-05T07:06:09.613798000+0100      74.53.140.153      25      10.10.1.4      1470S
: 235 Authentication succeeded
16      2009-10-05T07:06:09.614414000+0100      10.10.1.4      1470      74.53.140.153      25      C
: MAIL FROM: <gurpartap@patriots.in>
17      2009-10-05T07:06:09.956765000+0100      74.53.140.153      25      10.10.1.4      1470S
: 250 OK
18      2009-10-05T07:06:09.957250000+0100      10.10.1.4      1470      74.53.140.153      25      C
: RCPT TO: <raj_deol2002in@yahoo.co.in>
19      2009-10-05T07:06:10.319708000+0100      74.53.140.153      25      10.10.1.4      1470S
: 250 Accepted
20      2009-10-05T07:06:10.320203000+0100      10.10.1.4      1470      74.53.140.153      25      C
: DATA
21      2009-10-05T07:06:10.661679000+0100      74.53.140.153      25      10.10.1.4      1470S
: 354 Enter message, ending with "." on a line by itself
22      2009-10-05T07:06:10.692743000+0100      10.10.1.4      1470      74.53.140.153      25      C
: DATA fragment, 1460 bytes
23      2009-10-05T07:06:10.692786000+0100      10.10.1.4      1470      74.53.140.153      25      C
: DATA fragment, 1460 bytes
```

```

Session Actions Edit View Help
24 2009-10-05T07:06:10.692804000+0100 10.10.1.4 1470 74.53.140.153
: DATA fragment, 1460 bytes
25 2009-10-05T07:06:10.692823000+0100 10.10.1.4 1470 74.53.140.153 25 [
TCP Window Full] C: DATA fragment, 1460 bytes
26 2009-10-05T07:06:10.695115000+0100 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.
53.140.153 25 Destination unreachable (Fragmentation needed)
28 2009-10-05T07:06:10.695623000+0100 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.
53.140.153 25 Destination unreachable (Fragmentation needed)
29 2009-10-05T07:06:10.696248000+0100 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.
53.140.153 25 Destination unreachable (Fragmentation needed)
30 2009-10-05T07:06:10.696634000+0100 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.
53.140.153 25 Destination unreachable (Fragmentation needed)
32 2009-10-05T07:06:11.104972000+0100 10.10.1.4 1470 74.53.140.153 25 [
TCP Out-Of-Order] C: DATA fragment, 1452 bytes
33 2009-10-05T07:06:11.104998000+0100 10.10.1.4 1470 74.53.140.153 25 [
TCP Out-Of-Order] C: DATA fragment, 1452 bytes
35 2009-10-05T07:06:11.469759000+0100 10.10.1.4 1470 74.53.140.153 25 [
TCP Out-Of-Order] C: DATA fragment, 1452 bytes
36 2009-10-05T07:06:11.469814000+0100 10.10.1.4 1470 74.53.140.153 25 [
TCP Out-Of-Order] C: DATA fragment, 1452 bytes
38 2009-10-05T07:06:11.494181000+0100 10.10.1.4 1470 74.53.140.153 25 C

```

```

Session Actions Edit View Help
: DATA fragment, 1452 bytes
39 2009-10-05T07:06:11.494199000+0100 10.10.1.4 1470 74.53.140.153 25 C
: DATA fragment, 1452 bytes
41 2009-10-05T07:06:11.834628000+0100 10.10.1.4 1470 74.53.140.153 25 C
: DATA fragment, 1452 bytes
42 2009-10-05T07:06:11.834655000+0100 10.10.1.4 1470 74.53.140.153 25 C
: DATA fragment, 1452 bytes
44 2009-10-05T07:06:11.858316000+0100 10.10.1.4 1470 74.53.140.153 25 C
: DATA fragment, 1452 bytes
45 2009-10-05T07:06:11.858334000+0100 10.10.1.4 1470 74.53.140.153 25 f
rom: "Gurpartap Singh" <gurpartap@patriots.in>, subject: SMTP, (text/plain) (text/html) (tex
t/plain) (text/plain) (text/plain) | .
52 2009-10-05T07:06:12.248789000+0100 74.53.140.153 25 10.10.1.4 1470S
: 250 OK id=1Mugho-0003Dg-Un
54 2009-10-05T07:06:14.763825000+0100 10.10.1.4 1470 74.53.140.153 25 C
: QUIT
56 2009-10-05T07:06:15.105467000+0100 74.53.140.153 25 10.10.1.4 1470S
: 221 xc90.websitewelcome.com closing connection

(ponzhi@kali)-[~/SBT-DF203-Lab4]
$

```

Part B - Identify Commands and Response Codes

Sequence	Direction	Command/Code	Meaning	Timestamp
1	Server to Client	220	Service ready (xc90.websitewelcome.com ESMTP Exim 4.69)	07:06:08.219663
2	Client to Server	EHLO GP	Client greeting	07:06:08.224809
3	Server to Client	250	Capabilities: SIZE, PIPELINING, AUTH PLAIN LOGIN, STARTTLS, HELP	07:06:08.566183
4	Client to Server	AUTH LOGIN	Authentication method selected	07:06:08.568729
5	Client to Server	MAIL FROM	Envelope sender: <gurpartap@patriots.in>	07:06:09.614414
6	Client to Server	RCPT TO	Envelope recipient: <raj_deol2002in@yahoo.co.in>	07:06:09.957250
7	Client to Server	DATA	Message content begins (354 continue)	07:06:10.320203
8	Client to Server	QUIT	Session termination (221 closing connection)	07:06:14.763825

```

(ponzhi@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'smtp.req || smtp.rsp' -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e smtp.req.command -e smtp.req.parameter \
-e smtp.response.code -e smtp.rsp.parameter \
| tee reports/smtp_commands_responses.tsv
6      2009-10-05T07:06:08.219663000+0100      74.53.140.153      10.10.1.4      2
20      xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 ,We do not
      authorize the use of this system to transport unsolicited, ,and/or bulk e-mail.
7      2009-10-05T07:06:08.224809000+0100      10.10.1.4      74.53.140.153      EHLO      GP
9      2009-10-05T07:06:08.566183000+0100      74.53.140.153      10.10.1.4      2
50      xc90.websitewelcome.com Hello GP [122.162.143.157],SIZE 52428800,PIPELINING,AUTH PLAI
N LOGIN,STARTTLS,HELP
10     2009-10-05T07:06:08.568729000+0100      10.10.1.4      74.53.140.153      AUTH      LOGIN
11     2009-10-05T07:06:08.911081000+0100      74.53.140.153      10.10.1.4      3
34      VXNlcm5hbWU6
12     2009-10-05T07:06:08.911655000+0100      10.10.1.4      74.53.140.153
13     2009-10-05T07:06:09.253544000+0100      74.53.140.153      10.10.1.4      3
34      UGFzc3dvcmQ6
14     2009-10-05T07:06:09.254118000+0100      10.10.1.4      74.53.140.153

```

```
ponzhi@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help
14 2009-10-05T07:06:09.254118000+0100 10.10.1.4 74.53.140.153
15 2009-10-05T07:06:09.613798000+0100 74.53.140.153 10.10.1.4 2
35 Authentication succeeded
16 2009-10-05T07:06:09.614414000+0100 10.10.1.4 74.53.140.153 MAIL FROM:
<gurpartap@patriots.in>
17 2009-10-05T07:06:09.956765000+0100 74.53.140.153 10.10.1.4 2
50 OK
18 2009-10-05T07:06:09.957250000+0100 10.10.1.4 74.53.140.153 RCPT TO: <
raj_deol2002in@yahoo.co.in>
19 2009-10-05T07:06:10.319708000+0100 74.53.140.153 10.10.1.4 2
50 Accepted
20 2009-10-05T07:06:10.320203000+0100 10.10.1.4 74.53.140.153 DATA
21 2009-10-05T07:06:10.661679000+0100 74.53.140.153 10.10.1.4 3
54 Enter message, ending with "." on a line by itself
52 2009-10-05T07:06:12.248789000+0100 74.53.140.153 10.10.1.4 2
50 OK id=1Mugho-0003Dg-Un
54 2009-10-05T07:06:14.763825000+0100 10.10.1.4 74.53.140.153 QUIT
56 2009-10-05T07:06:15.105467000+0100 74.53.140.153 10.10.1.4 2
21 xc90.websitewelcome.com closing connection

(ponzhi@kali)-[~/SBT-DF203-Lab4]
```

```
ponzhi@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help

(ponzhi@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'frame.number==12' -T fields -e tcp.stream
0

(ponzhi@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,STREAM | tee reports/smtp_stream_STREAM.txt
tshark: Invalid address.
tshark: Invalid range.
tshark: Invalid parameter.

(ponzhi@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,0 | tee reports/smtp_stream_STREAM.txt

Follow: tcp,ascii
Filter: tcp.stream eq 0
Node 0: 10.10.1.4:1470
Node 1: 74.53.140.153:25
```

```
ponzhi@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help
(ponzhi@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,0 | tee reports/smtp_stream_Si
REAM.txt

Follow: tcp,ascii
Filter: tcp.stream eq 0
Node 0: 10.10.1.4:1470
Node 1: 74.53.140.153:25
181
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220-We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.

9
EHLO GP

137
250-xc90.websitewelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800
250-PIPELINING
```

```
ponzhi@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help
220 and/or bulk e-mail.

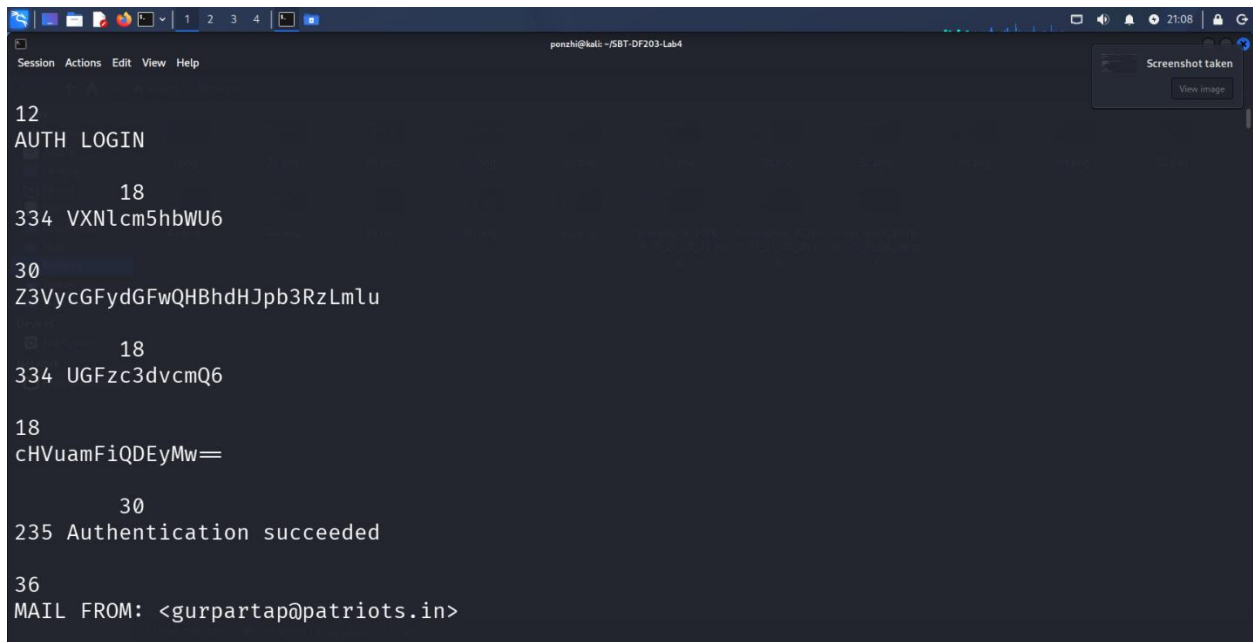
9
EHLO GP

137
250-xc90.websitewelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800
250-PIPELINING
250-AUTH PLAIN LOGIN
250-STARTTLS
250 HELP

12
AUTH LOGIN

18
334 VXNlcm5hbWU6

30
Z3VycGFydGFwQHBhdHJpb3RzLmLu
```



A terminal window titled "ponchi@kali: ~/SBT-DF203-Lab4" with a menu bar (Session, Actions, Edit, View, Help) and a toolbar (1, 2, 3, 4). The terminal displays an SMTP session log. A "Screenshot taken" notification is visible in the top right corner.

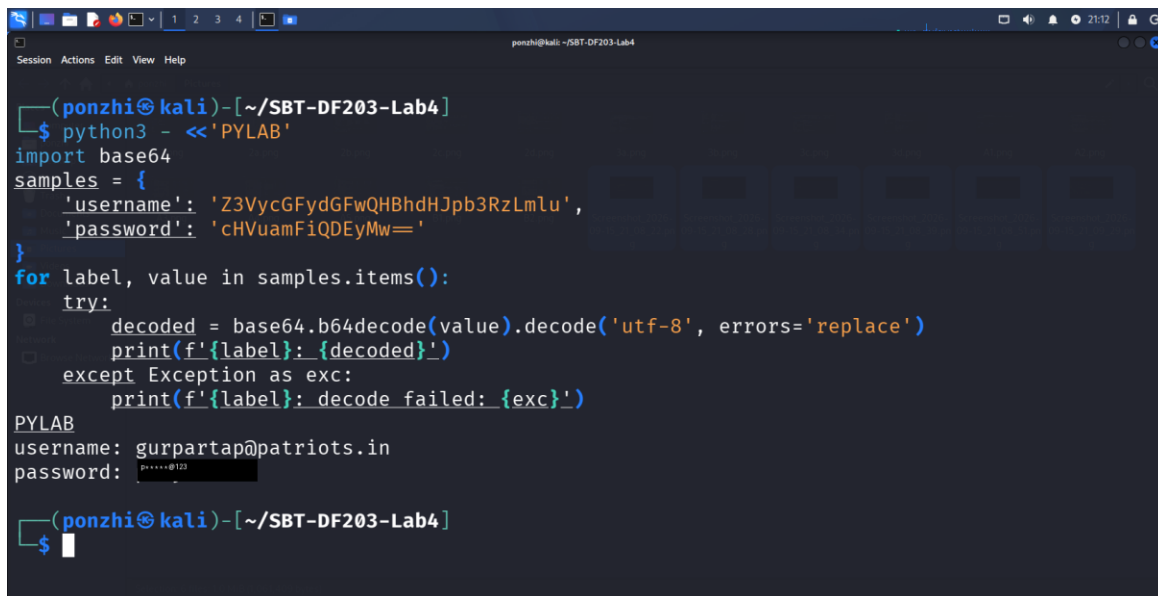
```
12 AUTH LOGIN
18
334 VXNlcm5hbWU6
30
Z3VycGFydGFwQHBhdHJpb3RzLmLu
18
334 UGFzc3dvcmQ6
18
cHVuamFiQDEyMw==
30
235 Authentication succeeded
36
MAIL FROM: <gurpartap@patriots.in>
```

Part C - Decode Base64 Authentication Evidence

Base64 is a way to encode data, but it is not encryption. The AUTH LOGIN credentials found in plain SMTP traffic were decoded offline using Python's base64 module. No online decoder was used, which meets the lab requirements..

Captured Base64 values from the AUTH LOGIN exchange:

Field	Captured Base64 Value
Username (frame 12)	Z3VycGFydGFwQHBhdHJpb3RzLmlu
Password (frame 14)	cHVuamFiQDEyMw==

A screenshot of a terminal window with a dark background. The prompt is '(ponzhi@kali) - [~/SBT-DF203-Lab4]'. The user enters '\$ python3 - <<'PYLAB''. The script imports base64 and defines a dictionary 'samples' with 'username' and 'password' keys. It then iterates over the items, decoding each value using base64.b64decode and decoding the result with 'utf-8'. The output shows 'username: gurpartap@patriots.in' and 'password: pu*****23'.

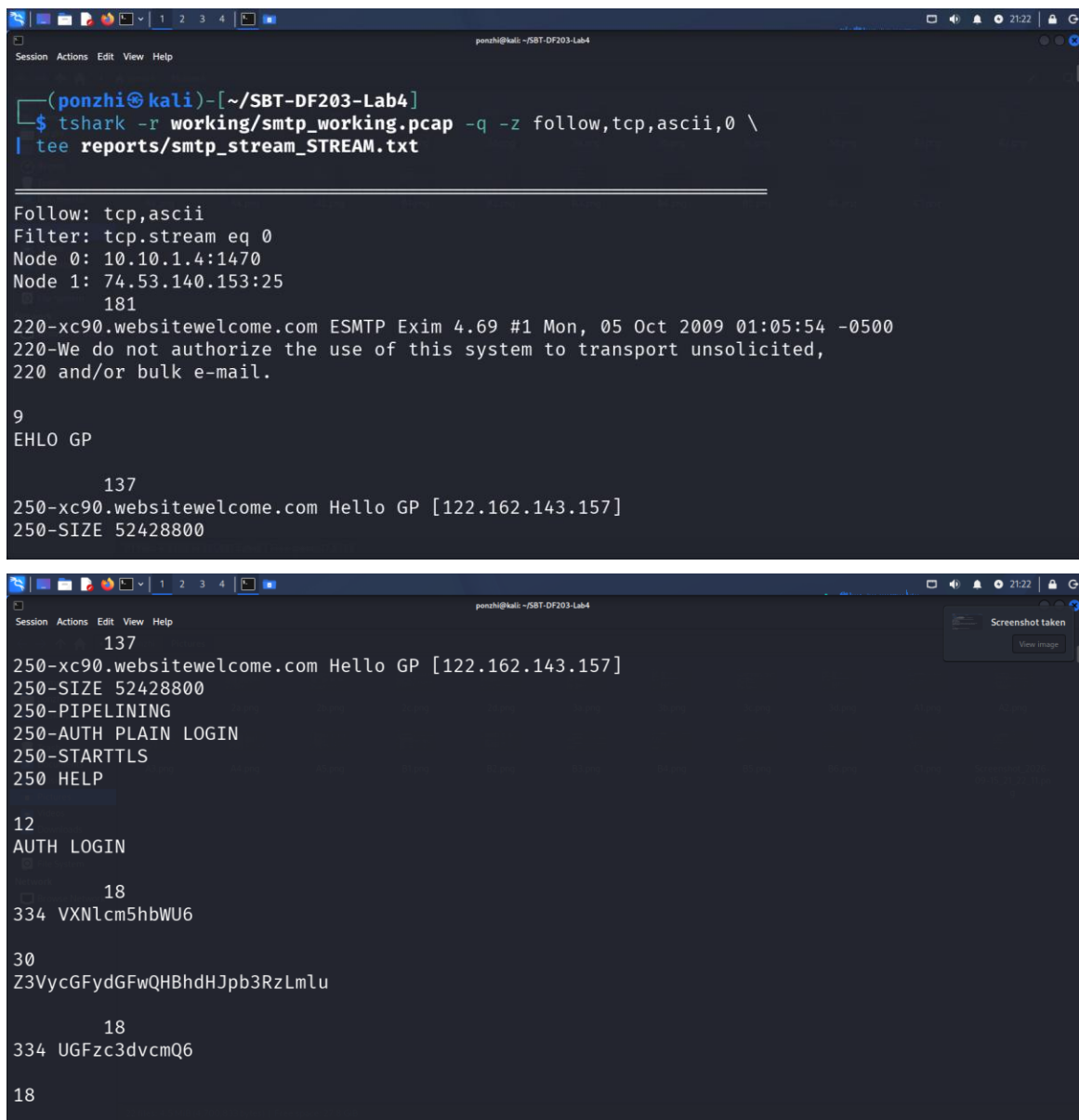
```
(ponzhi@kali) - [~/SBT-DF203-Lab4]
$ python3 - <<'PYLAB'
import base64
samples = {
    'username': 'Z3VycGFydGFwQHBhdHJpb3RzLmlu',
    'password': 'cHVuamFiQDEyMw=='
}
for label, value in samples.items():
    try:
        decoded = base64.b64decode(value).decode('utf-8', errors='replace')
        print(f'{label}: {decoded}')
    except Exception as exc:
        print(f'{label}: decode failed: {exc}')
PYLAB
username: gurpartap@patriots.in
password: pu*****23
(ponzhi@kali) - [~/SBT-DF203-Lab4]
$
```

Decoded and redacted result, as required for the public-facing report body.

Field	Masked / Redacted Value
Username	gurpartap@patriots.in
Password	pu*****23

Part D - Reconstruct the Email Message

The TCP stream (tcp.stream eq 0) was followed in ASCII mode to reconstruct the full application-layer conversation, from the initial connection through message delivery.



```
ponzhi@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help

(ponzhi@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,0 \
| tee reports/smtp_stream_STREAM.txt

Follow: tcp,ascii
Filter: tcp.stream eq 0
Node 0: 10.10.1.4:1470
Node 1: 74.53.140.153:25
181
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220-We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.

9
EHLO GP

137
250-xc90.websitewelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800

137
250-xc90.websitewelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800
250-PIPELINING
250-AUTH PLAIN LOGIN
250-STARTTLS
250 HELP

12
AUTH LOGIN

18
334 VXNlcm5hbWU6

30
Z3VycGFydGFwQHBhdHJpb3RzLmlu

18
334 UGFzc3dvcmQ6

18
```

```
Session Actions Edit View Help
AUTH LOGIN

18
334 VXNlcm5hbWU6

30
Z3VycGFydGFwQHBhdHJpb3RzLmlu

18
334 UGFzc3dvcmQ6

18
cHVuamFiQDEyMw==

30
235 Authentication succeeded

36
MAIL FROM: <gurpartap@patriots.in>

8
```

Message headers extracted from the DATA section of the reconstructed stream.

```
Session Actions Edit View Help
ponzhi@kali: ~/SBT-DF203-Lab4

(ponzhi@kali)-[~/SBT-DF203-Lab4]
$ grep -Ei '^(Date|From|To|Subject|Message-ID|MIME-Version|Content-Type|User-Agent|X-Mailer):' \
reports/smtp_stream_STREAM.txt | tee reports/message_headers.txt
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
X-Mailer: Microsoft Office Outlook 12.0
Content-Type: multipart/alternative;
Content-Type: text/plain;
Content-Type: text/html;
Content-Type: text/plain;

(ponzhi@kali)-[~/SBT-DF203-Lab4]
$
```

Header	Value
From	"Gurpartap Singh" <gurpartap@patriots.in>
To	<raj_deol2002in@yahoo.co.in>
Subject	SMTP
Date	Mon, 5 Oct 2009 11:36:07 +0530
Message-ID	<000301ca4581\$ef9e57f0\$cedb07d0\$@in>
MIME-Version	1.0
Content-Type	multipart/mixed.
X-Mailer	Microsoft Office Outlook 12.0

Body Type and Attachment Assessment: The Content-Type of multipart/mixed shows that the message contains multiple parts, which often means there could be an attachment included with the message text. The captured data reveals text content divided into two segments with 47 and 12 lines, indicating a multipart setup. However, this analysis did not fully retrieve the individual MIME part headers or any attachment filenames. This is noted as a limitation in Section 9, rather than a confirmed finding. To determine if a file attachment is present, a complete export of the data section with MIME part boundaries would be needed.

Part E - Determine Client, Hosts and Network Metadata

Frame-level metadata was extracted to positively identify the endpoints at both the network and data-link layers.

```
(ponzhi@kali)~[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
-e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e tcp.srcport \
-e ip.dst -e tcp.dstport -e tcp.stream \
| tee reports/smtp_network_metadata.tsv
6      1254722768.219663000    00:1f:33:d9:81:60    00:e0:1c:3c:17:c2    74.53.140.153
25     10.10.1.4      1470    0
7      1254722768.224809000    00:e0:1c:3c:17:c2    00:1f:33:d9:81:60    10.10.1.4    1
470    74.53.140.153    25      0
9      1254722768.566183000    00:1f:33:d9:81:60    00:e0:1c:3c:17:c2    74.53.140.153
25     10.10.1.4      1470    0
10     1254722768.568729000    00:e0:1c:3c:17:c2    00:1f:33:d9:81:60    10.10.1.4    1
470    74.53.140.153    25      0
11     1254722768.911081000    00:1f:33:d9:81:60    00:e0:1c:3c:17:c2    74.53.140.153
25     10.10.1.4      1470    0
12     1254722768.911655000    00:e0:1c:3c:17:c2    00:1f:33:d9:81:60    10.10.1.4    1
470    74.53.140.153    25      0
13     1254722769.253544000    00:1f:33:d9:81:60    00:e0:1c:3c:17:c2    74.53.140.153
25     10.10.1.4      1470    0
14     1254722769.254118000    00:e0:1c:3c:17:c2    00:1f:33:d9:81:60    10.10.1.4    1
```

```
(ponzhi@kali)~[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'smtp contains "User-Agent" || smtp contains "X-Mailer"' \
-T fields -e frame.number -e tcp.stream -e smtp.req.parameter -e data-text-lines \
| tee reports/client_indicators.tsv
22     0      Line-based text data (47 lines)
26     0      Line-based text data (12 lines)

(ponzhi@kali)~[~/SBT-DF203-Lab4]
$
```

Endpoint	Details
Client	IP 10.10.1.4, port 1470, MAC 00:1f:33:d9:81:60
Server	IP 74.53.140.153, port 25, MAC 00:e0:1c:3c:17:c2
TCP stream index	0

Part F - Encryption and Evidential Limitations

- ✓ The server indicated that it supports STARTTLS, which allows for upgrading to a secure connection, in its EHLO response. However, the client did not request a STARTTLS command and went straight to AUTH LOGIN over an unsecured channel. As a result, there was no TLS handshake in this packet capture, and everything—commands, authentication details, and message content remains visible and recoverable.
- ✓ Just knowing the port number (25) does not confirm that the traffic was unencrypted. The conclusion is based on the lack of a STARTTLS command and the absence of a TLS handshake in the captured data.
- ✓ If STARTTLS had been used correctly, only minimal details would be visible afterward, like IP addresses, ports, timing, and certificate information—not the actual commands, credentials, or message text.
- ✓ Another issue noted was that there were temporary ICMP "Destination unreachable (Fragmentation needed)" messages coming from an intermediate host (192.168.1.1) and some TCP packets arrived out of order during DATA transfer. This is related to the way the network is set up, not a security problem, but it did affect how we analyzed the data.

Required Findings Worksheet

Question	Finding
When did the SMTP session start and end?	Started 2009-10-05 07:06:08.22 (220 banner); ended 07:06:15.
Client IP/MAC and port	10.10.1.4, MAC 00:1f:33:d9:81:60, port 1470
Server IP/MAC and port	74.53.140.153, MAC 00:e0:1c:3c:17:c2, port 25
SMTP server banner	220-xc90.websiteswelcome.com ESMTP Exim 4.69 1 Mon, 05 Oct 2009 01:05:54 -0500
Client software	Microsoft Office Outlook 12.0
Authentication method	AUTH LOGIN Base64-encoded username/password exchanged in plaintext
Envelope sender and recipient	MAIL FROM: <gurpartap@patriots.in>; RCPT TO: <raj_deol2002in@yahoo.co.in>
Message From/To/Subject	From: "Gurpartap Singh" <gurpartap@patriots.in>; To: <raj_deol2002in@yahoo.co.in>; Subject: SMTP
Message body type	multipart/mixed
Attachment present?	The analysis does not provide a clear answer based only on the headers. The multipart/mixed structure indicates it might be possible, but we could not confirm any attachment filenames or MIME parts in this review.
STARTTLS/TLS observed?	The server offered the option for a secure connection using STARTTLS, but the client did not use it. As a result, the entire session was not secure and remained in plain text.
Key limitations	This time, no TLS was used, so all the content was fully recoverable. If STARTTLS had been used, only the metadata would be available. There were also transient ICMP messages about fragmentation needed and TCP.

Conclusions

This analysis recreated a clear text SMTP session between an Outlook 12.0 client and an Exim mail server. It included the AUTH LOGIN credential exchange, routing information, and message headers. The offline Base64 decoding proved that the authenticated identity matched the claimed sender of the message. No encryption was used, even though the server offered STARTTLS. This means that the entire session is recoverable from the captured data alone. One

question remains: did the message include a file attachment? To answer this, further extraction of the DATA section is needed, so it is marked for follow-up rather than assumed.

Appendix - Protected Evidence (Restricted Distribution)

This section contains sensitive values that are kept only for instructor and protected submission purposes, according to the lab's evidence-handling rules. Do not share this information in public discussion channels or in unredacted report copies.

Field	Full Decoded Value
Username	gurpartap@patriots.in
Password	punjab@123

Academic Integrity and Authorization Statement

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations, and conclusions.